

DHYEY PATEL

Embedded Systems & Firmware

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Experience

Robotics Production Intern

Jun 2026 - Aug 2026

Electrans Technologies Ltd.

Oakville, Ontario

- Designed an automated wire-sleeving feed system in **SolidWorks**, sizing drum, shaft, and drive components to index preset lengths directly to the cutter, replacing manual measurement while cutting cycle time by **60%** and improving length tolerance by **50%**.
- Identified supplier machining defects in pneumatic brake-line fittings during incoming QA and escalated the nonconformance to engineering, preventing approximately **\$10,000 in defective components** and avoiding brake-line leaks, rework, and field failures.
- Performed electrical, pneumatic, and dimensional QA using DMM continuity checks, GD&T drawings, calipers, go/no-go gauges, and leak tests to catch wiring faults, out-of-tolerance features, machining defects, and assembly nonconformances before release.
- Helped triple weekly output from about **100 to 300 finished receptacles** by rotating independently across assembly and QA stations, balancing workload, and improving task sequencing, jig setup, inspection flow, and overall production workflow.

Firmware Team Member

Sep 2025 - Dec 2025

University of Waterloo Design Team (Electrium Mobility)

Waterloo, Ontario

- Integrated **BMS telemetry over UART in C++** into a reusable interface for battery monitoring, diagnostics, and control.
- Compared battery-pack configurations across capacity, mass, packaging, and electrical limits, supporting a revised design that increased tested top speed by **4 km/h** while meeting mounting, clearance, and integration constraints during prototyping.
- Resolved BMS/controller mounting, cable-routing, and connector-access constraints before prototype integration testing.

Engineering Intern

Jul 2024 - Aug 2024

Tulsar Controls

Brantford, Ontario

- Fabricated, terminated, and crimped **1,000+ wire leads** for industrial control systems, then wired and labeled control panels from electrical schematics to customer specifications while verifying conductor routing and terminal placement during QA.
- Verified panel wiring and component placement against schematics, catching assembly discrepancies before final electrical testing.

Projects

Autonomous Object-Fetching Robot

Aug 2026 - Present

C++, ESP-IDF, FreeRTOS, Linux, Raspberry Pi 5, ESP32, UART, TCP/IP, Encoders, PID, Python, ROS 2

- Developed ESP32 firmware in **C++ with ESP-IDF and FreeRTOS** for encoder feedback, PWM motor control, and closed-loop PID wheel-speed regulation, reducing straight-line drift by **65% over 2 m tests** while maintaining stable differential-drive motion.
- Built a multithreaded **C++ Linux service on Raspberry Pi 5** to exchange wheel setpoints and telemetry with the ESP32 over UART and expose rover commands/status over TCP/IP, linking high-level software to the real-time motor-control layer.
- Automated testing with Python scripts to send repeatable velocity commands, log telemetry to CSV, and compare commanded vs measured wheel speed, using the results to tune PID before integrating ROS 2 velocity commands for a simple indoor drive routine.

STM32 Motor Control Test Bench

Sep 2026

STM32F072RB, C, SPI, ADC, PWM, TIM1, STM32 HAL, STM32CubeIDE

- Built an STM32 motor-control test bench that converts potentiometer input through an external SPI ADC to proportional PWM, **cutting motor-speed and servo-range characterization time by 80%** versus repeated firmware edits and reflashes.
- Configured and debugged SPI, chip-select GPIO, TIM1 PWM, and ADC communication with STM32 HAL, then verified the end-to-end control path on an oscilloscope for reliable transfers, PWM timing, and proportional duty-cycle response.

Autonomous Recycling Sorter

Sep 2025 - Dec 2025

C++, Sensors, Motors, Embedded Control, 3D Printing

- Programmed an autonomous recycling sorter in **C++** using calibrated reflectivity sensing and motorized routing, achieving **90% sorting success across 40 trials at 15 items/min** during repeated testing under varied lighting and item placement.
- Tuned sensor thresholds, motor timing, and routing sequences through repeated testing to reduce jams and false classifications, maintaining stable sorting under changing lighting, item placement, and sensor noise without manual recalibration between trials.

Technical Skills

Programming and Development: C, C++, Python, Git, Linux/Ubuntu, Docker, CMake

Embedded and Controls: STM32, ESP32, Raspberry Pi, SPI, UART, I2C, ADC, PWM, Timers, PID Control, FreeRTOS

Robotics and Debugging: ROS 2, STM32CubeIDE, STM32 HAL, OpenCV, Multimeter, Oscilloscope, Soldering, Crimping

Education

University of Waterloo

Sep 2025 - Present

Bachelor of Applied Science in Mechatronics Engineering

Waterloo, Ontario

Completed Coursework: Circuits, Data Structures and Algorithms, Statics, Engineering Design and Communication

Coursework in Progress: Digital Logic and PLCs, Differential Equations, Dynamics, Mechanics of Deformable Solids